

Evidence Integrity Validator

Forensic File Integrity Validator

User Manual - Version 1.0

For Windows | Works offline | No installation required

Table of Contents

- 1. Overview
- 2. Installation
- 3. Quick Start Guide
- 4. Features Explained
- 5. File Hashing
- 6. Metadata Extraction
- 7. Hash Verification
- 8. Report Generation
- 9. Troubleshooting
- 10. License & Support

1. Overview

The Evidence Integrity Validator is a forensic tool for digital investigators, law enforcement, and cybersecurity experts.

- * Compute cryptographic hashes (MD5, SHA1, SHA256, SHA512, BLAKE2b)
- * Extract metadata from images, PDFs, and Office documents
- * Verify files against known hash values
- * Generate court-ready PDF reports with chain of custody
- * Export results to CSV

Everything works offline - no internet required. Your evidence never leaves your machine.

2. Installation

System Requirements:

Requirement	Details
OS	Windows 10/11
RAM	512 MB (2 GB rec.)
Disk	150 MB
Internet	Not required

Installation Steps:

Step 1: Download EvidenceValidator.exe

Step 2: Double-click to run

Step 3: If SmartScreen warns, click More info then Run anyway

Step 4: Browser opens at <http://localhost:8081>

3. Quick Start Guide

Step 1 - Drop Files: Drag files into the dashed box or click browse.

Step 2 - Compute Hashes: Select algorithm and click Compute Hashes.

Step 3 - Review Results: See hash values, file size, and date.

Step 4 - Generate Report: Click Generate Report (PDF).

4. Features Explained

Area	Description
Case Info	Reference, name, agency
Drop Files	Drag-and-drop upload
Loaded Files	Session files
Actions	Hash, verify, report
Results	Hashes and details
Metadata	EXIF, GPS, author

5. File Hashing

Hashing creates a unique digital fingerprint of a file.

Algorithm	Length	Use
MD5	32	Legacy
SHA-1	40	Legacy
SHA-256	64	Recommended
SHA-384	96	High sec.
SHA-512	128	Max sec.
BLAKE2b	128	Modern

6. Metadata Extraction

- * Images: Camera model, GPS coordinates, date taken
- * PDFs: Author, creation/modification dates
- * Office docs: Author, revision number, company

7. Hash Verification

Step 1: Load and hash your file

Step 2: Paste the expected hash

Step 3: Click Verify - shows MATCH or MISMATCH

8. Report Generation

PDF reports include: case info, chain of custody, file details, hash values, verification results, metadata, and examiner certification.

9. Troubleshooting

- * SmartScreen warning: Click More info then Run anyway
- * Browser issue: Open <http://localhost:8081> manually
- * Large files: Allow time for files over 1GB

10. License & Support

- * Trial: 3 files per session
- * Professional: Unlimited files
- * Enterprise: Custom branding
- * Email: ibbiy@icloud.com